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LETTER

Detection and attribution of extreme precipitation changes over India

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Abstract

Extreme precipitation events are intensifying across India, posing significant risks to its population and economy. While observations indicate widespread changes in rainfall extremes across the country, the role of different climate drivers remains poorly quantified, particularly at regional scales. This study provides the first comprehensive detection and attribution (D&A) of changes in extreme precipitation events across India's homogeneous monsoon regions. Using the regularised optimal fingerprinting D&A framework, gridded daily rainfall observations from India Meteorological Department, and a 14-member multi-model ensemble from the Coupled Model Intercomparison Project Phase 6, we disentangle the contributions of greenhouse gases (GHGs), anthropogenic aerosols, and natural forcings to the observed changes in extreme precipitation. Observed extreme indices—Rx1day, Rx5day, R95p and R99p—were examined over India and found to exhibit contrasting trends across the homogeneous monsoon zones, with notable increases in the core monsoon region (West central (WC)), a weaker trend over the southern peninsula, and a marked decline over Northeast (NE) India in recent decades. D&A analysis was carried out for Annual and monsoon season extreme indices over homogeneous monsoon zones using decadal-average time series, and provides robust evidence that human activities have already altered India's rainfall extremes with anthropogenic fingerprints detected across most regions and indices. GHG forcing emerges as the dominant driver of intensifying precipitation extremes, most consistently detected and attributed in the WC and Central NE regions, which make up the core monsoon zone, while anthropogenic aerosols exert a substantial counteracting influence, particularly since the mid-20th century. These results are consistent with previous findings of their effects on mean monsoon precipitation across global monsoon regions, and particularly the South Asian monsoon. Our findings suggest that the apparent absence of a strong increase in extreme precipitation does not indicate weak climate sensitivity, but highlight the competing effects of GHG-induced intensification and aerosol-driven suppression, with important implications for future risk as aerosol emissions decline.

1. Introduction

India's climate is shaped by the South Asian monsoon, marked by strong seasonal and spatial rainfall variability. While important for water, agriculture, and the economy, the monsoon also drives extreme precipitation events—ranging from floods and flash floods to landslides that cause major socioeconomic losses. India's developing economy and dense population heightens its vulnerability, underscoring the need to understand how these extremes are changing in a warming climate.

There is observational evidence that rainfall over India has intensified (Sen Roy and Balling 2004, Goswami *et al* 2006, Rajeevan *et al* 2008, Mishra and Liu 2014, Roxy *et al* 2017, Kumar *et al*

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2021, Kumari and Kumar 2022, Pant *et al* 2024). Goswami *et al* (2006) reported rising frequency and magnitude of extreme rainfall events alongside a decline in moderate rains over central India (1951–2000), while Rajeevan *et al* (2008) linked increasing extremes to warming tropical Indian Ocean sea surface temperatures.

Detection and attribution (D&A) studies help disentangle the relative contributions of anthropogenic forcings, such as greenhouse gases (GHGs) and aerosols, from natural variability. Mukherjee *et al* (2018) attributed the rise in Annual maximum daily rainfall (Rx1day) over India during 1951–2015 to anthropogenic warming but did not consider the effects of aerosols and GHGs individually. A large body of evidence identifies anthropogenic aerosols as the dominant driver of the post-1950 weakening of the global monsoons. Ramanathan *et al* (2005) and Chung and Ramanathan (2006) established the role of black carbon in suppressing monsoon rainfall and masking surface warming, thereby weakening the North Indian Ocean meridional SST gradient. Bollasina *et al* (2011, 2014) confirmed that widespread South Asian drying was due to aerosol-driven suppression of the tropical meridional overturning circulation, with local emissions having the strongest precipitation-reducing effect. Undorf *et al* (2018) showed that the decline cannot be explained without anthropogenic aerosols, with remote North American/European emissions dominating before 1975 and local South Asian emissions thereafter. Polson *et al* (2014) showed that aerosols were responsible for the decline in Northern Hemisphere monsoon rainfall from the 1950s to the mid-1980s.

While further advances have been made in attributing changes in globally averaged precipitation (Zhang *et al* 2007, Marvel and Bonfils 2013) and over the Indian region (Bollasina *et al* 2011, 2014, Sharma *et al* 2021), attributing observed changes in heavy precipitation to human influence remains a significant challenge, especially at the regional level. Progress is hindered by the high natural variability of rainfall, limitations in the quality and length of observational records, and the inability of climate models to realistically simulate intense, localised precipitation (Stott *et al* 2010, Sarojini *et al* 2016). Studies focusing on the Northern Hemisphere, including regional studies over North America, have attributed the intensification of Annual maximum 1 day (Rx1day) and 5 day (Rx5day) precipitation to anthropogenic forcing (Zhang *et al* 2013, Kirchmeier-Young and Zhang 2020). Studies by Dong *et al* (2020) and Paik *et al* (2020) have identified a clear human contribution to increases in percentile-based precipitation extremes globally, with GHGs as the dominant driver and aerosols providing a partial offsetting effect. Over China and the Asian monsoon region, Dong *et al* (2022), Xu *et al* (2022), Chen *et al* (2024); have attributed anthropogenic causes for the intensification of rainfall.

Studies on extreme precipitation over India are mainly focused on trends and variability, but the formal D&A of observed trends remains unexplored. This study addresses the gap by applying D&A techniques to long-term extreme precipitation indices across India. Quantifying the relative contributions of anthropogenic and natural forcings helps determine whether observed trends are likely to persist under continued emissions or primarily reflect internal variability. This distinction has important implications for mitigation strategies aimed at reducing long-term risk, as well as for adaptation planning and resilient infrastructure. In section 2, we describe the data and methods used in the study. Results are presented in section 3, followed by conclusions in section 4.

2. Data and methods

2.1. Observational data and study region

Daily gridded rainfall data from the India Meteorological Department (IMD) at $1^\circ \times 1^\circ$ resolution for the period 1905–2014 (Rajeevan *et al* 2006) were utilised for this study. Extreme precipitation indices were calculated following the definitions of Zhang *et al* (2011) and Donat *et al* (2013). Indices computed include Rx1day, Rx5day (Maximum 1 day and consecutive 5 day precipitation, respectively), R95p, and R99p (total precipitation from days exceeding the 95th and 99th percentiles of daily precipitation, respectively). These indices are calculated for the Annual and monsoon JJAS seasons at each grid point.

Our study focuses on the homogeneous monsoon rainfall regions defined as in Parthasarathy *et al* (1993). These regions are distinguished by consistent seasonal rainfall characteristics driven by monsoonal dynamics and include the Northwest (NW), Northeast (NE), Central NE (CNE), West Central (WC; Core Monsoon Zone), South Peninsula (SP), and hilly regions (see supplementary figure S1). Hilly regions were excluded from the analysis due to known observational inconsistencies and sparse station coverage in these areas, which can introduce uncertainties in gridded rainfall data (Kothawale and Rajeevan 2017). Whereas the homogeneous zones were demarcated based on monsoon season rainfall, the extremes of rainfall also follow a similar spatial distribution as seen in figure 1. Values of the extreme rainfall indices in the grid points within a homogeneous region are then spatially averaged to arrive at an Annual or JJAS time series for each region from which decadal averages are created.

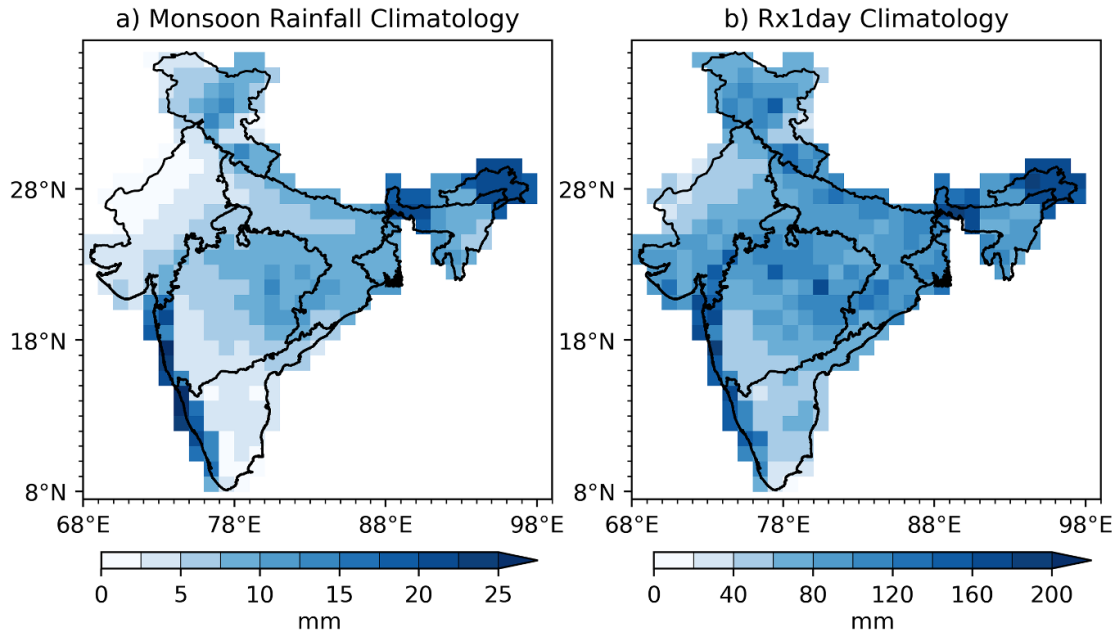


Figure 1. (a) Monsoon daily rainfall climatology and (b) Rx1day climatology for the monsoon (JJAS) season over India, computed for the period 1905–2014 using the IMD 1° × 1° gridded dataset. The overlaid boundaries denote the homogeneous monsoon regions of India.

2.2. Model data

We utilise daily precipitation from the DAMIP (Gillett *et al* 2016) experiments that are part of the Coupled Model Intercomparison Project Phase 6 (CMIP6; Eyring *et al* 2016), specifically the individual forcing GHG only (hist-GHG), natural forcing only (hist-Nat), and anthropogenic aerosol only (hist-Aer) forcing simulations in addition to the historical and piControl simulations. A total of 14 CMIP6 models common to the four historical experiments are considered in this study, with multiple ensemble members available for each model and forcing experiment (see table S1). Of these, 12 models provide piControl runs. Additionally, piControl runs from 12 more CMIP6 models were included to have a better diversity of climate models used to estimate noise in the D&A analysis.

Extreme indices for the Annual and monsoon seasons were first computed on each model's native grid and subsequently regridded to the common 1° × 1° observational grid using the conservative regridding method (Jones 1999). We then compute individual model ensemble means, followed by the multi-model ensemble mean for each experiment, from which the sub-regional series were obtained by area averaging.

2.3. Methods

2.3.1. Regularised optimal fingerprinting (ROF)

ROF (Ribes *et al* 2013) is an extension of the widely used optimal fingerprinting methodology for climate change D&A (Allen and Tett 1999, Tett *et al* 2002, Allen and Stott 2003). ROF provides a more objective and statistically efficient D&A framework by addressing limitations of classical optimal fingerprinting, which typically projects climate data onto leading empirical orthogonal functions (EOFs) to estimate internal climate variability but may introduce sensitivity to EOF truncation and reduce attribution accuracy (Ribes *et al* 2013, Hannart *et al* 2014). Its effectiveness has been demonstrated in various applications, including precipitation attribution studies (Hannart *et al* 2014, Li *et al* 2023) and D&A of temperature changes over India (Dileepkumar *et al* 2018).

In the ROF method, the observations Y are regressed onto n model simulated signals X_i representing the responses of extreme precipitation indices to individual external forcings (e.g. GHGs, anthropogenic aerosols, and natural forcings). The regression equation for the total least squares (TLS) model can be written as,

$$Y = \sum_{i=1}^{n} (X_i - \varepsilon_{X_i})\beta_i + \varepsilon$$

where Y represents the observed Annual/JJAS extreme precipitation index (e.g. Rx1day) over the period 1905–2014, ε represents the regression residual, i.e. The component of the observations not explained by the linear combination of the forcing-specific response patterns, while ε_{Xi} represents the sampling uncertainty in each model-simulated response pattern arising from finite ensemble size, and β_i is the regression coefficient or the scaling factor.

We apply a two-signal regression model to separate the influences of anthropogenic (Ant) and natural (Nat) forcings and two three-signal models as implemented in Dileepkumar *et al* (2018) (with natural (Nat), GHGs and other anthropogenic forcings (OA)) and in the studies by Dong *et al* (2021) and Xu *et al* (2022) (with GHGs, aerosols (Aer), and natural (Nat) forcings). The ROF is applied to area-averaged precipitation indices aggregated as decadal mean time series over each homogeneous monsoon region of India. For the All-India (ALLIN) analysis, we construct an area-weighted aggregate time series from the regionally averaged time series for the five homogeneous regions so that it represents the area-integrated response across the five homogeneous regions rather than a simple arithmetic average. Further details of the data preparation and use of piControl runs for noise estimation can be found in the supplementary text.

3. Results

3.1. Observed trends in extreme precipitation

We first examine changes in extreme precipitation indices derived from the IMD ($1^\circ \times 1^\circ$) gridded rainfall dataset. Figure 2 presents the pentadal and decadal mean time series of the Rx1day, Rx5day, R95p and R99p indices for both Annual and JJAS seasons. Annual values generally exceed JJAS values, suggesting contributions from non-monsoon extremes, particularly in the SP region. The SP region (and to a lesser extent the NE) shows that the Annual and JJAS time series are separated in magnitude, indicating the importance of the other months; for example, the post-monsoon for SP when tropical cyclones and the following NE monsoon can bring extreme rainfall.

The evolution of extremes over the 110 years reveals regionally distinct patterns. All indices show marked increases in most regions during the latter half of the 20th century. The WC region (part of the core monsoon zone) exhibits strong increases since the 1940s. The NW region also exhibits a long-term increase, but with large inter-decadal variability. In contrast, the SP region shows weaker trends and exhibits large inter-decadal variability. The NE region shows increasing trends up to the 1980s, followed by strong decreasing trends in all the extreme indices, unlike in other regions.

The trend values calculated for each of the decadal and pentadal average time series (see table S2) indicate that WC has seen statistically significant increases in almost all indices, with NW, NE and ALLIN showing increases in the Rx1day, R95p and R99p indices. The R99p index indicates statistically significant increases in all the regions—except SP, which interestingly, shows no significant increases in any of the indices. Taken together, these results provide observational evidence of long-term intensification of precipitation extremes across core monsoon regions, most notably WC. By evaluating both pentadal and decadal averages, we highlight robust persistent trends over and above short-term fluctuations, providing a solid observational basis for a D&A analysis.

3.2. Comparison of observation and model simulations

In order to have confidence that the models are fit for the purpose of carrying out a D&A analysis, it is important to evaluate the ability of the CMIP6 models to simulate the fundamental characteristics of extreme precipitation over the study region. We compare linear trends and interannual variability of the four key extreme precipitation indices against observations from the IMD for the period 1905–2014. Figure 3 presents the linear trend (abscissa) against the standard deviation for individual ensemble members from the 14 CMIP6 models for the Annual indices over the 1905–2014 period (the corresponding figure for JJAS values is shown in the supplementary material figure S2). Observed trends are represented by black bars, with 5%–95% confidence intervals calculated as described in Santer *et al* (2000).

Rx1day and Rx5day reveal a notable consistency between the observations and the model ensemble across most regions (figures 3(a) and (b)). The SP region reveals a few outliers (BCC-CSM2-MR, FGOALS-G3, IPSL-CM6A-LR)—models with too high variability and trend compared to observations. Similarly, in the WC region, most models underestimate the trend, except where models such as CanESM5, CESM2, and IPSL-CM6A-LR show good agreement with the observed trend. The most prominent feature of the percentile-based indices (figures 3(c) and (d)) is the substantially larger spread across models in the variability and trends (see for instance, R95p over NE). Within this large spread, some models like HadGEM3-GC31-LL and MRI-ESM2-0 simulate a level of interannual variability comparable to the IMD observation. Individual models clearly have a tendency to simulate lower variability in certain regions (for example, IPSL-CM6A-LR in the CNE and NE regions, but not the other regions for all 4 indices). The NE and WC regions of India show large positive trends in observed precipitation extremes, a feature that CESM2 and the ACCESS family models generally capture.

A key similarity across all regions is that the IMD-derived trend value is almost never a complete outlier as verified by a two-sided empirical rank-based test (not shown), which lends confidence to using this model ensemble. It is also important to note that no single model consistently performs well across all metrics and regions. A comparison between the JJAS

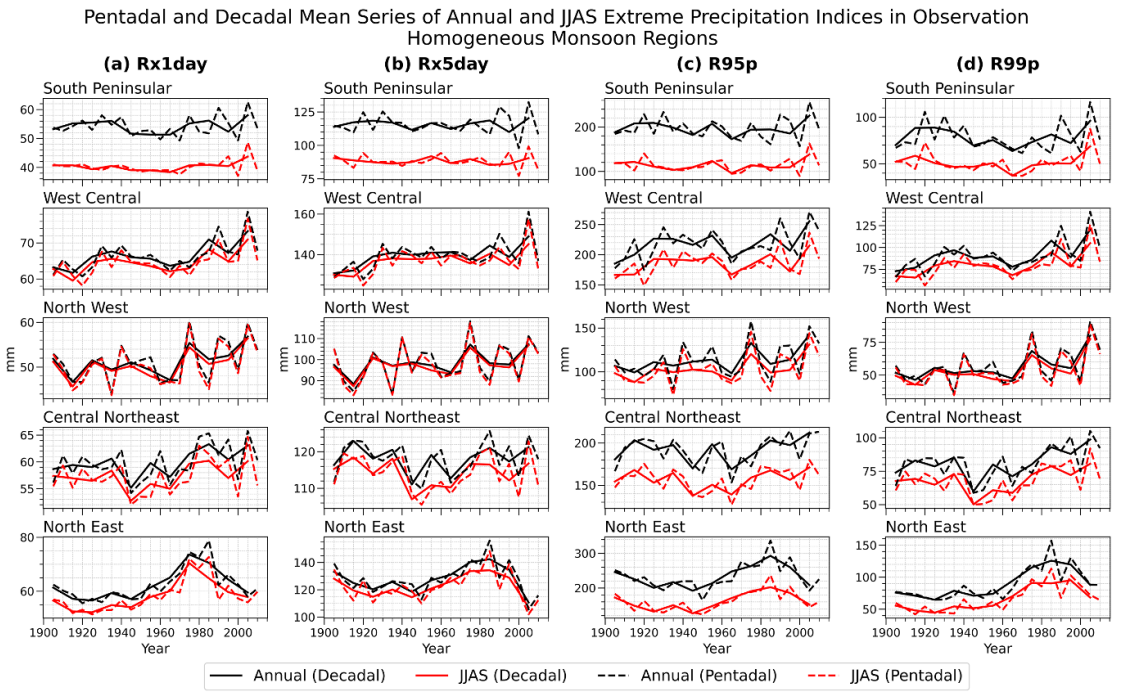


Figure 2. Decadal (solid lines) and pentadal (dashed lines) average time series of extreme precipitation indices for (a) Rx1day, (b) Rx5day, (c) R95p, and (d) R99p across five homogeneous monsoon regions of India for the period 1905–2014 for Annual (black) and JJAS (red) season. from the IMD 1° × 1° gridded daily rainfall dataset. Note the different vertical scales in the different panels.

(figure 3) and Annual statistics (figure S2) shows that the fundamental model-observation relationships are consistent across the two seasons. Importantly for the D&A analysis, the simulated variability is usually greater than the observed variability across all regions and indices (except perhaps Rx1day in the NE region, where only a few of the models capture the magnitude of the observed variability). This evaluation demonstrates that the CMIP6 models perform well in capturing the behaviour of heavy rainfall events, with specific models showing strong agreement with observations in different regions and are therefore fit for use in carrying out a D&A analysis.

3.3. Extreme precipitation in individual forcing simulations

To assess the relative contribution of different external forcings on extreme precipitation, we analyse decadal mean time series of four indices derived from CMIP6 historical (hist-All) and individual-forcing simulations (hist-GHG, hist-Aer, hist-Nat) across the 5 homogeneous monsoon regions (figure 4 presents the Annual time series with the corresponding JJAS series shown in figure S3). GHG-only simulations exhibit pronounced increases in all indices across all 5 regions, particularly after the mid-20th century. This upward trend is consistent with what is expected from GHG-driven warming and enhanced atmospheric moisture holding capacity. The strongest and most consistent increasing signals are found in the WC and NW regions.

In contrast, hist-Aer simulations show a declining or at best flat trend in extremes across the regions. This dampening effect is most evident in the WC and CNE regions as well as NE India, consistent with the known role of aerosol forcing in suppressing convection and monsoon rainfall (Ramanathan *et al* 2001, Bollasina *et al* 2011, Krishnan *et al* 2016). The hist-Aer simulations show a pronounced decreasing trend from the 1940s (and in some regions from the 1920s) to the 1970s, with a flattening out in subsequent years in all the regions (except the WC region, where recent decades have shown a further decrease in rainfall). The SP and NW regions exhibit a declining trend until the 1970s, followed by an increasing trend in rainfall, resulting in an overall rainfall change that appears flat. Natural-only forcing simulations (hist-Nat) exhibit low-amplitude variability without sustained trends, pointing to a limited role in explaining observed long-term intensification.

The hist-All simulations capture the combined effects of anthropogenic and natural drivers, yielding trends that typically fall between hist-GHG and hist-Aer scenarios, but with post-1970s increases aligning more closely with the GHG signal. The NE region

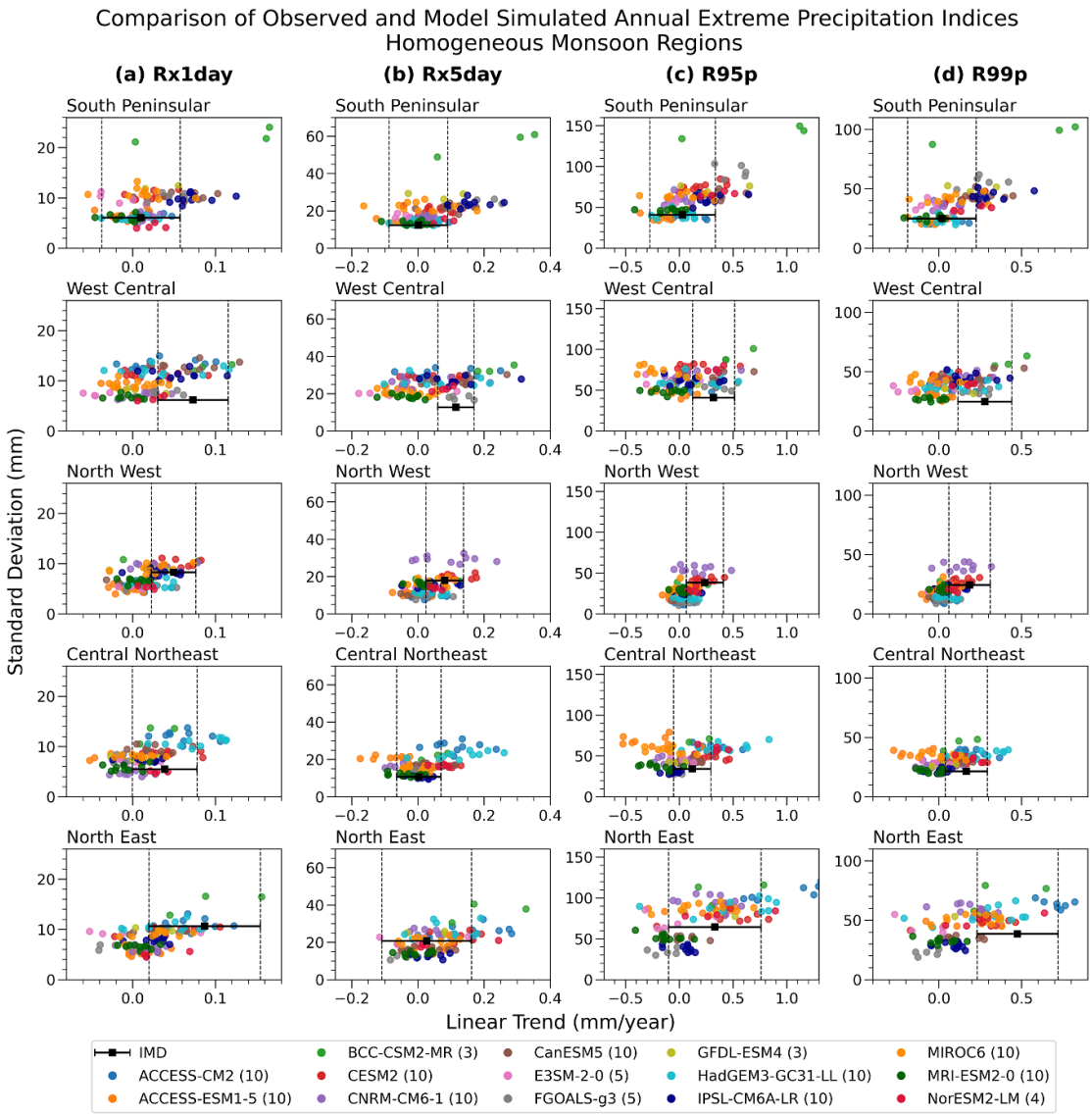


Figure 3. Comparison of observed and simulated linear trends and standard deviations of Annual extreme precipitation indices from CMIP6 historical simulations. The linear trend of the index time series is shown plotted on the abscissa against the standard deviation of the Annual time series of the extreme precipitation indices from 1905–2014. The four indices—(a) Rx1day, (b) Rx5day, (c) R95p, and (d) R99p (columns) are shown for five homogeneous monsoon regions over India (rows); South Peninsular (SP), West Central (WC), Northwest (NW), Central Northeast (CNE), and Northeast (NE). The black square represents the IMD observational dataset, with the horizontal error bar and dashed vertical lines indicating the 95% confidence interval of the observed trend. Coloured circles represent individual realisations of the 'historical' simulation from 14 CMIP6 models.

emerges as a hotspot where GHG-induced intensification is most evident, with R95p and R99p showing steep increases in recent decades, while the SP and NW regions show weaker signals.

Comparing across regions and indices, there is a high degree of consistency: increases in Rx1day and Rx5day are systematically accompanied by parallel increases in percentile-based indices, reinforcing the interpretation that anthropogenic GHGs are driving a broad redistribution of rainfall toward heavier extremes, rather than isolated intensification. Overall, GHG forcing appears to be the dominant driver of long-term increases in extreme precipitation indices, while aerosols act to offset part of this signal. Natural forcings alone do not result in any increases, emphasising the role of anthropogenic drivers in recent changes. The strong coherence among regions, indices, and temporal scales strengthens the motivation and provides a clear basis for carrying out the D&A analysis.

3.4. D&A

To quantify the contributions of external forcings to observed changes in regional precipitation extremes, we employ the ROF method for D&A across homogeneous monsoon regions. The observed time series of precipitation extremes were regressed onto model-simulated responses to individual forcings, with internal variability estimated from pre-industrial control simulations (piControl). A signal is considered detected if its scaling factor is distinct from zero and is considered attributed when its scaling

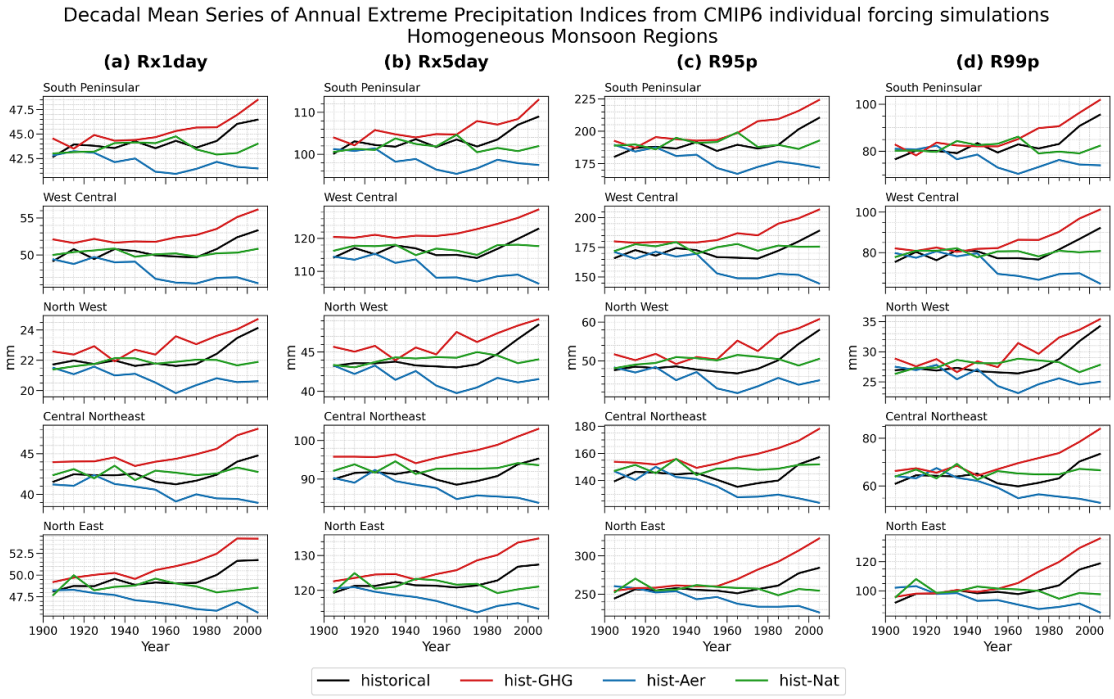


Figure 4. The Annual decadal mean extreme precipitation indices from CMIP6 individual forcing simulations for (a) Rx1day, (b) Rx5day, (c) R95p, and (d) R99p across five homogeneous monsoon regions of India for the period 1905–2014. The time series represent the multi-model ensemble (MME) mean response to different historical forcings: all forcings (hist-All, black), aerosol-only (hist-Aer, blue), greenhouse gas-only (hist-GHG, red), and natural-only (hist-Nat, green).

factor is consistent with unity (i.e. the confidence interval includes one). Scaling factors above one can also be considered attributed; assuming all confounding factors were accounted for, it implies a bias in the model response. Any signal with a confidence interval that crosses or is entirely below the zero line is considered not detected, signifying a lack of spatio-temporal consistency between the model fingerprints and observations.

Figure 5(a) shows the scaling factors for the two-signal analysis with uncertainties for Annual season and each region under decadal averaging (JJAS season shown in figure S4), with asterisks (*) marking the attributed cases (scaling factors consistent with unity) and plus (+) signs indicating attributed cases where the scaling factor is above one. The ant signal is attributed in the WC and CNE regions for the percentile indices and scaling factors in the NW indicating possible biases in model response. The influence of Nat is seen clearly in the WC, CNE and NW regions. It is notable that the NE, SP and ALLIN regions do not exhibit any detected or attributed cases.

The further decomposition of the ant signal can be performed in multiple ways; Dong *et al* (2021) and Xu *et al* (2022), Dileepkumar *et al* (2018), and Gillett *et al* (2021). Considering the Indian region and its high aerosol loading, we choose to first explore the decomposition into 3-signals, GHG, Aer and Nat as in Dong *et al* (2021) and Xu *et al* (2022) (supplementary text); the scaling factors of which are presented in figure 5(b).

We first note that, other than the NE and NW regions, most of the scaling factors for the four indices in all other regions indicate that a signal has been detected. In the NW region, the results highlight the challenge of detection in regimes dominated by high natural variability and detection is limited to GHG being attributed for Rx5day. Observed changes here remain indistinguishable from internal climate noise. In the SP region, both GHG and Aer forcing are attributed for Rx1day in the Annual and JJAS seasons (JJAS season shown in supplementary figure S4), indicating competing influences on short-duration rainfall intensity. The Annual Rx5day and R95p show attribution to GHG forcing, while R99p is attributed to Aer in both Annual and JJAS seasons.

The CNE region exhibits a strong anthropogenic influence, with concurrent attribution of GHG and Aer seen in Rx5day and R95p for Annual and monsoon seasons, as well as for Annual Rx1day. Even where the scaling factors are above one (marked with + symbol), the values are close enough to discount any serious model biases. The competing GHG-driven intensification and Aer-induced suppression is confirmed by the reconstructed time series (supplementary figure S5). This region has also seen large increases in sulphate aerosol loading in recent years (Li *et al* 2017) that could be playing a role in the rainfall suppression. This suggests that future changes in extreme precipitation over the CNE will depend on the balance between aerosol emissions and ongoing warming.

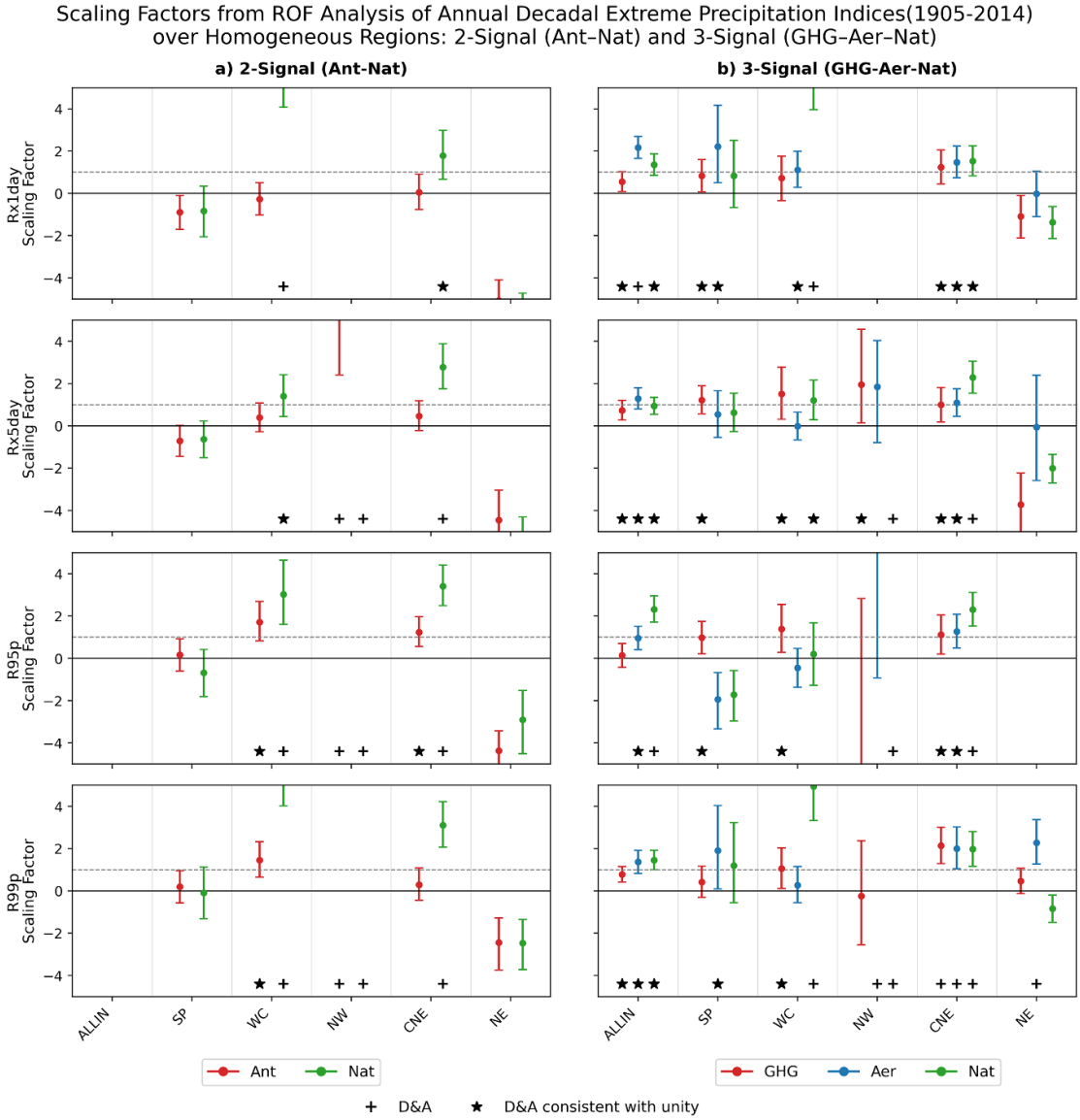


Figure 5. Results of the (a) two-signal (Ant, Nat) and (b) three-signal (GHG, Aer, Nat) ROF analysis for Annual extreme precipitation indices from 1905–2014 using Decadal (10 year) averaged time series. Scaling factors and their 95% confidence intervals are shown for GHG (red), Aer (blue), and Nat (green) forcings. Results are presented for four precipitation indices (rows) across different spatial scales: All-region combined(ALLIN) and five homogeneous monsoon regions (SP, WC, NW, CNE, NE). Asterisks (*) denote attributed signals, where the observed change is formally consistent with the model-simulated response to the respective forcing and Plus (+) signs indicate detected but not formally attributed signals.

In the WC region, GHG is detected or attributed for most indices at the Annual scale, with the notable exception of Rx1day, which is attributed to aerosol forcing in both Annual and JJAS seasons. Both R95p and R99p exhibit consistent GHG attribution across both Annual and monsoon seasons, while Annual Rx5day also shows GHG attribution. These patterns suggest a dominant role of GHG in shaping the extremes in WC, alongside a more limited aerosol influence on short-duration rainfall intensity. The reconstructions (figure S5) indicate that the GHG signal has risen sharply since 1970, while the aerosol masking effect is weaker. Consequently, the GHG-induced intensification is emerging as the dominant driver.

At the ALLIN scale, GHG, Aer and Nat forcings are attributed for Rx5day (Annual) and R99p (Annual) signals. For the Rx1day (Annual), GHG and Nat are attributed, while for R95p (Annual), only Aer is attributed. Fewer attribution cases emerge for the JJAS season. For Annual and JJAS indices, both GHG and Aerosol signals (and in many cases, Nat) are consistently detected. This consistency confirms that the anthropogenic fingerprints are distinguishable from internal variability.

Attribution in the NE to any of the forcings is sparse, with aerosol forcing attributed only to Annual R99p. The scaling factors for GHG are negative, signifying a lack of spatio-temporal consistency between the model fingerprints and observations.

This suggests that the regression framework cannot physically attribute the observed trends in this region to GHG forcing, likely due to the dominance of regional dynamical processes or internal variability that are not captured by the models. While the observations show a decreasing trend in extremes from the 1990s–2014, the raw model fingerprints generally predict increased extremes due to GHGs. This indicates that the fingerprints fail to capture the specific regional dynamics driving the NE's recent rainfall decline. The discrepancy over the NE region is likely due to a combination of the timing of internal variability modes and missing forcings. Internal multi-decadal variability (e.g. Pacific decadal oscillation and Atlantic multidecadal oscillation) has been identified in literature as one likely factor responsible for the region's recent drying trend (Joshi and Rai 2015, Choudhury *et al* 2019). The modes of variability and their timing and magnitude are something that CMIP6 model simulations used would not be able to replicate from observations (Deser *et al* 2012, Fasullo *et al* 2020, Du and Chen 2024).

We now address the issue of missing climate forcings in the model simulations or in the D&A analysis. Since our 3-signal analysis includes only GHG, Aer and Nat—which is an incomplete subset of the full CMIP6 forcings, we perform another 3-signal analysis where the climate signals are decomposed as GHG, OA and Nat (details in supplementary text). The signals attributed from this analysis (GHG + OA + Nat; figure S6) are nearly identical to the ones from the GHG + Aer + Nat analysis, with the same regions (WC, CNE and ALLIN) emerging with a clear message of GHG along with either Aer or OA influence. The NE remains a region where changes are unexplained by the larger set of anthropogenic forcings included in OA. This points to the potential omission of localised forcings in CMIP6 models. One such missing driver is intensive agricultural irrigation, which has been shown in idealised model simulations to decrease mean rainfall over the NE region (Mathur and AchutaRao 2019). Given that very few model simulations with irrigation exist, this hypothesis will require new simulations and analysis to investigate.

Taken together, these results demonstrate that historical changes in extreme precipitation over India are controlled by region-dependent interplay between anthropogenic and natural forcings. Aerosols emerge as the dominant suppressing influence on extremes at the ALLIN scale, effectively masking the GHG-driven intensification. At the regional scale, strong contrasts are evident, with some regions exhibiting clear competition between GHG warming and aerosol cooling, while others show only weak signals. The additional role of natural variability further highlights the complexity of regional extreme precipitation responses. We examine trends reconstructed using the best estimate scaling factors for individual forcings GHG, Aer and Nat in each of the regions (figure S7) with particular attention to the WC and CNE regions with most of the attributed signals. It is seen that the observed and reconstructed trends are comparable and where the signs do not match, are very close to zero. The reconstructed trends clearly indicate the compensation of GHG-induced positive trends by Aer-induced negative trends. For the ALLIN region, the reconstructed trends are less reliable as they include regions where all the signals have not been attributed. It is worth noting that the recent methodological work has highlighted potential undercoverage of confidence intervals derived from TLS, particularly in weak-signal or highly collinear regimes (DelSole *et al* 2018, Li *et al* 2021, Ma *et al* 2023). The possibility of undercoverage implies that the reported confidence intervals should be interpreted with caution, especially in regions where detection is marginal. Overall, the results indicate that the absence of a strong observed increase in extreme precipitation over India does not reflect weak climate sensitivity, but rather the compensating effects of multiple forcings acting across different spatial and temporal scales.

4. Conclusions

This study presents the first D&A analysis of long-term changes in extreme precipitation across India. Using gridded observations from IMD between 1905–2014, DAMIP simulations from CMIP6, and the ROF method, we estimate the contributions of GHGs, aerosols, and natural forcings to changes in rainfall extremes.

We first evaluated CMIP6 models' ability to simulate the extreme precipitation indices (Rx1day, Rx5day, R95p, and R99p) over India's monsoon zones. CMIP6 models forced with historical forcings reproduce the observed trends and variability in all four extreme indices over most regions. While the observed trends in extreme precipitation indices over the 1905–2014 period are increasing over most regions (and especially in the WC region that forms part of the core monsoon zone), the NE region is unique in producing a decreasing trend in recent decades—something that is not captured in CMIP6 models. This could be due to a mismatch in the timing of multi-decadal variability modes between models and observations or possibly missing climate forcings.

We then carried out the D&A analysis using the ROF method. A two-signal analysis separating anthropogenic (Ant) and natural (Nat) forcings detects anthropogenic influence in the WC, CNE,

and NW regions. The three-signal analysis where the GHG, Aer and Nat signals are used reflects the compensating effects of GHGs and anthropogenic aerosols over India, where GHG forcing intensifies precipitation extremes while aerosols exert a competing regional drying influence. Combining these opposing responses into a single anthropogenic signal dampens the net forced response, making the three-signal configuration more effective for detecting anthropogenic influence against internal climate variability. To investigate whether any of the OA signals resulting from the full set of CMIP6 forcings (that include ozone and land-use land cover changes etc) has an influence, we performed a second three-signal analysis separating GHG, OA forcings, and natural (Nat) signals. The resulting scaling factors are broadly consistent with those obtained from the primary GHG-Aer-Nat analysis. The GHG signal remains positive and frequently detectable across regions and indices, while the OA signal exhibits magnitudes comparable to the aerosol signal in several regions, indicating that anthropogenic aerosols dominate the non-GHG anthropogenic contribution to extreme precipitation changes over India. Our results provide evidence of a significant human fingerprint on India's rainfall extremes, yielding several key conclusions: First, that GHG forcing is the dominant driver of the observed increases in precipitation extremes. The GHG signal is robustly detected and attributed across most regions; Second, that anthropogenic aerosols have exerted a significant dampening effect, historically suppressing precipitation extremes; third, our analysis reveals that the detection of GHG and Aer influences on extreme rainfall is seen predominantly in the WC and CNE regions (where in recent decades the effect of Aer in weakening the extreme rainfall is overwhelmed by increases brought on by GHGs); finally, at the ALLIN scale, both Annual and JJAS extremes are characterised by a strong attribution of GHG-induced increases as well as suppression by aerosols. This combination of signals explains the long-standing absence of a strong observed increase in rainfall intensity over India, demonstrating that the muted trends primarily reflect substantial masking of GHG-driven intensification rather than a weak thermodynamic response. These results are consistent with previous research, which has established that anthropogenic aerosols have a dampening effect on mean monsoon precipitation over the global monsoon regions, particularly the South Asian monsoon, masking a portion of the intensification driven by GHGs (Bollasina *et al* 2011, Myhre *et al* 2017). This counteracting influence is often detected as a strong but regionally variable signal in formal attribution studies (Undorf *et al* 2018, IPCC 2021). The detection of these two opposing anthropogenic signals underscores the complex human influence on the monsoon system.

The implications of these findings are significant. The robust attribution of intensifying extremes to GHG emissions provides a strong scientific basis for climate mitigation policies and reinforces the urgency of the global energy transition. Furthermore, the powerful masking effect of aerosols presents a critical challenge for future adaptation in India. As India and the surrounding region continue to improve air quality and reduce aerosol pollution, the suppressive effect on rainfall may diminish, potentially unmasking the full force of GHG-driven intensification and leading to an accelerated increase in flood risk as well as increased temperature extremes. This necessitates that regional adaptation and infrastructure planning account for a future where rainfall and temperature extremes may be even more severe than seen in historical trends.

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Data availability statement

The gridded daily rainfall dataset used in this study is provided by the India Meteorological Department (IMD) Data Supply Portal, Climate Research and Services Pune and is available at www.imdpune.gov.in/cmpg/Griddata/. CMIP6 model simulations were obtained from the Earth System Grid Federation (ESGF) archive and are openly available at https://pcmdi.llnl.gov/CMIP6/. The shapefiles for the homogeneous summer monsoon regions of India used in this study are available on GitHub at https://github.com/Cassimsannan/Shapefiles/tree/master.

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